

# Engineering Chemistry-1

**CHE109**

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# Electromagnetic Radiation

How the energy travel trough the space

- The light from the sun
- The energy used to cook food in a microwave oven
- infrared light,
- ultraviolet light,
- X-rays and  $\gamma$ -radiations

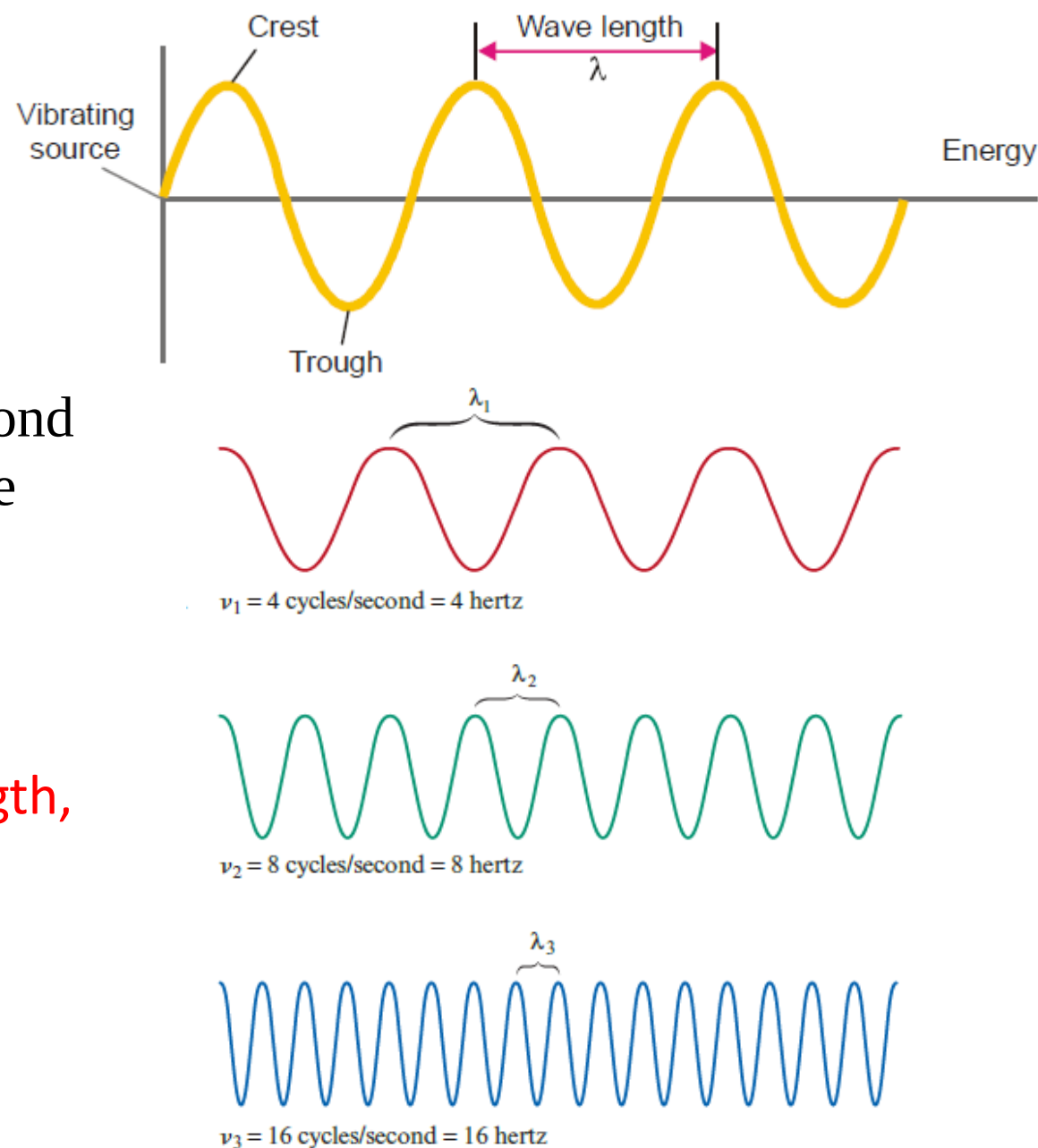
Although these forms of radiant energy seem quite different, they all exhibit the same type of wavelike behavior and travel at the speed of light in a vacuum.

**Wavelength:** the distance between two successive crests or troughs of a wave and denoted by the Greek letter  $\lambda$  (lambda).

Unit: centimeters or meters or in *angstrom*

**Frequency:** the number of waves (cycles) per second that pass a given point in space. Symbolized by the lowercase Greek letter nu,  $\nu$ .

a wave of high frequency has a shorter wavelength,  
while a wave of low frequency has a longer  
wavelength.



**The speed (or velocity)** of a wave is the distance through which a particular wave travels in one second.

$$c = f\lambda$$

or Speed = Frequency  $\times$  Wavelength

Speed is denoted by  $c$  and it is expressed in cm per second.

## Wave Number

This is reciprocal of the wavelength and is given the symbol  $\bar{\nu}$  (nu bar). That is,

$$\bar{\nu} = \frac{1}{\lambda}$$

The wave number is the number of wavelengths per unit of length covered. Its units are  $\text{cm}^{-1}$  or  $\text{m}^{-1}$ .

*M1. The brilliant red colors seen in fireworks are due to the emission of light with wavelengths around 650 nm when strontium salts such as  $\text{Sr}(\text{NO}_3)_2$  and  $\text{SrCO}_3$  are heated. (This can be easily demonstrated in the lab by dissolving one of these salts in methanol that contains a little water and igniting the mixture in an evaporating dish.) Calculate the frequency of red light of wavelength  $6.50 \times 10^2$  nm.*

**Solve:** We can convert wavelength to frequency using the equation

$$c = f \times \lambda$$

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m s}^{-1}}{6.5 \times 10^{-7} \text{ m}} = 4.61 \times 10^{14} \text{ s}^{-1} = 4.61 \times 10^{14} \text{ Hz}$$

*M2: The laser in an audio CD player uses light with a wavelength of  $7.80 \times 10^2$  nm. Calculate the frequency of this light.*

*M3: What is the frequency of violet light with a wavelength of 408 nm?*

*M4: Light with a wavelength of 478 nm lies in the blue region of the visible spectrum. Calculate the frequency of this light.*

*M5: The frequency of the strong red line in the spectrum of potassium is  $3.91 \times 10^{14}/s$ . What is the wavelength of this light in nanometers?*

Solve: We can convert wavelength to frequency using the equation

$$c=f \times \lambda$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m s}^{-1}}{3.91 \times 10^{14} \text{ s}^{-1}} = 7.67 \times 10^{-7} \text{ m} = 767 \times 10^{-9} \text{ m} = 767 \text{ nm}$$

*M6: Radio waves in the AM region have frequencies in the range 550 to 1600 kilocycles per second (550 to 1600 kHz). Calculate the wavelength corresponding to a radio wave of frequency  $1.365 \times 10^6/s$  (that is, 1365 kHz).*

*M7: Microwaves have frequencies in the range 109 to 1012/s (cycles per second), equivalent to between 1 gigahertz and 1 terahertz. What is the wavelength of microwave radiation whose frequency is  $1.258 \times 10^{10}/s$ ?*

# Do It Your self

1. The wavelength of the green light from a traffic signal is centered at 522 nm. What is the frequency of this radiation?
2. What is the wavelength (in meters) of an electromagnetic wave whose frequency is  $3.64 \times 10^{-7}$  Hz.
3. (a) What is the frequency of light having a wavelength of 456 nm? (b) What is the wavelength (in nanometers) of radiation having a frequency of  $2.45 \times 10^{-9}$  Hz? (This is the type of radiation used in microwave ovens.)
4. The average distance between Mars and Earth is about  $1.3 \times 10^8$  miles. How long would it take TV pictures transmitted from the Viking space vehicle on Mars' surface to reach Earth? (1 mile = 1.61 km.)
5. How many minutes would it take a radio wave to travel from the planet Venus to Earth? (Average distance from Venus to Earth = 28 million miles.)

# From Classical Physics to Quantum Theory

The classical before nineteenth-century got limited success to understand atoms and molecules.

Assuming molecules behaves like rebounding balls can explain few macroscopic phenomena (such as the pressure exerted by a gas)

Could not explain stability of molecules; the forces that hold atoms together.

The properties of atoms and molecules are not governed by the same physical laws as larger objects.

## Max Planck the game changer



### Older Idea

Energy is continuous and that any amount of energy could be released in a radiation process.

Radiation emitted by solids heated to various temperatures



### New Idea

**atoms and molecules emit energy only in certain discrete quantities, or *quanta*.**

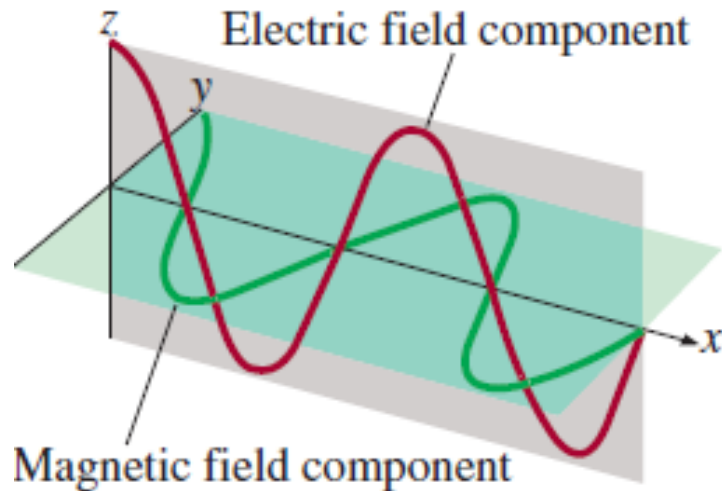


Altered our concept of nature forever.

# Electromagnetic Radiation

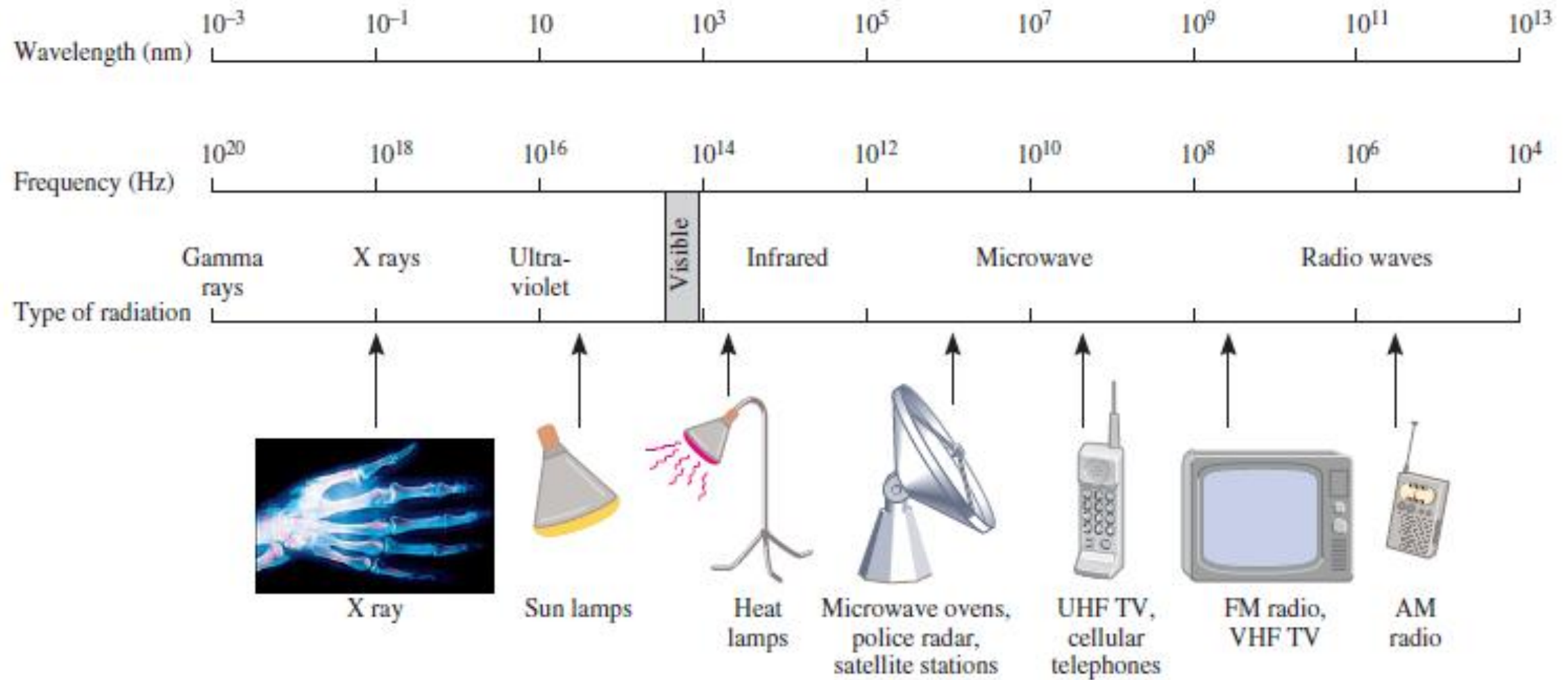
Electromagnetic wave has an electric field component and a magnetic field component. These two components have the same wavelength and frequency, and hence the same speed, but they travel in mutually perpendicular planes .

Sound waves and water waves are not electromagnetic waves, but X rays and radio waves are.

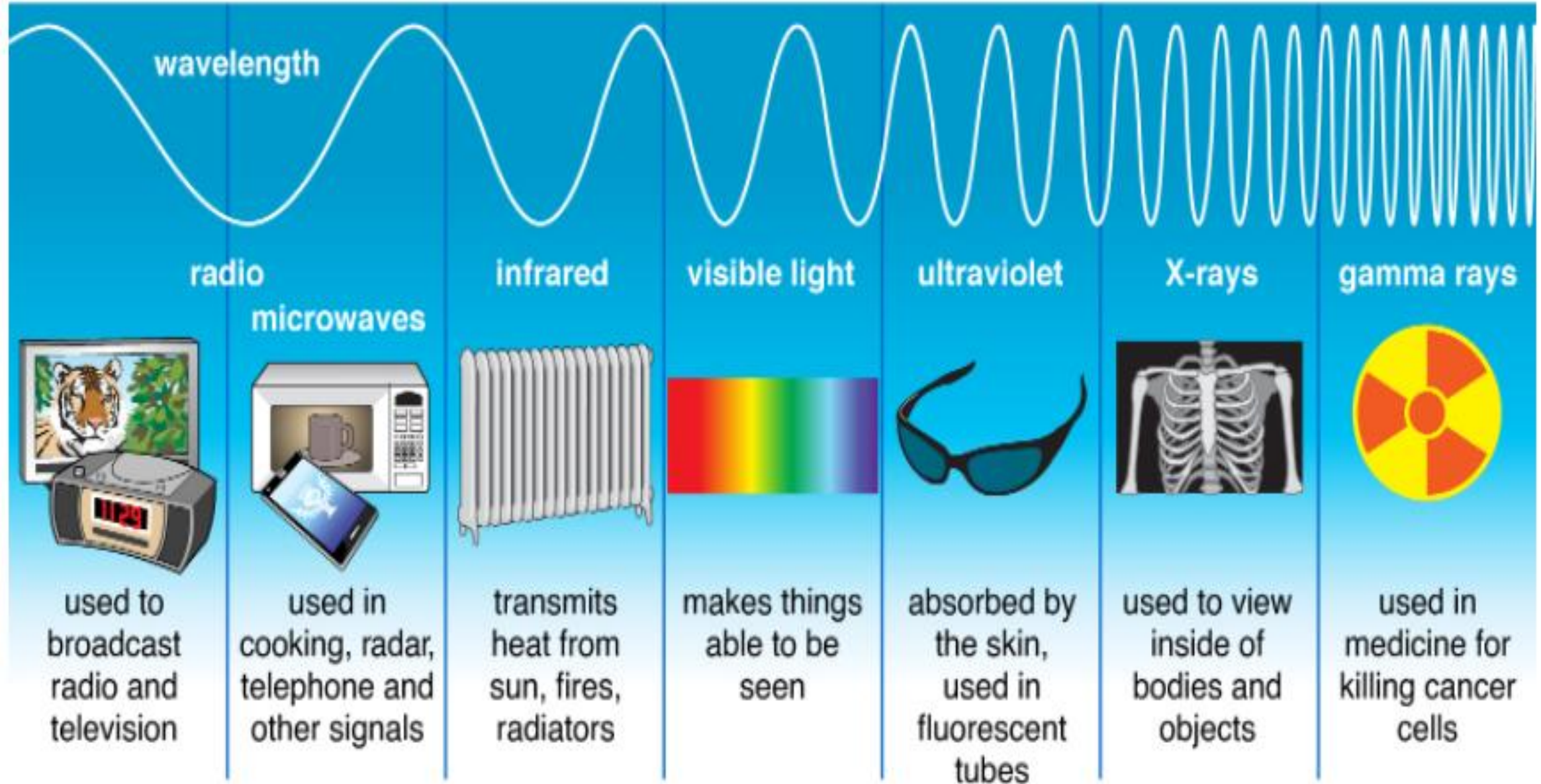


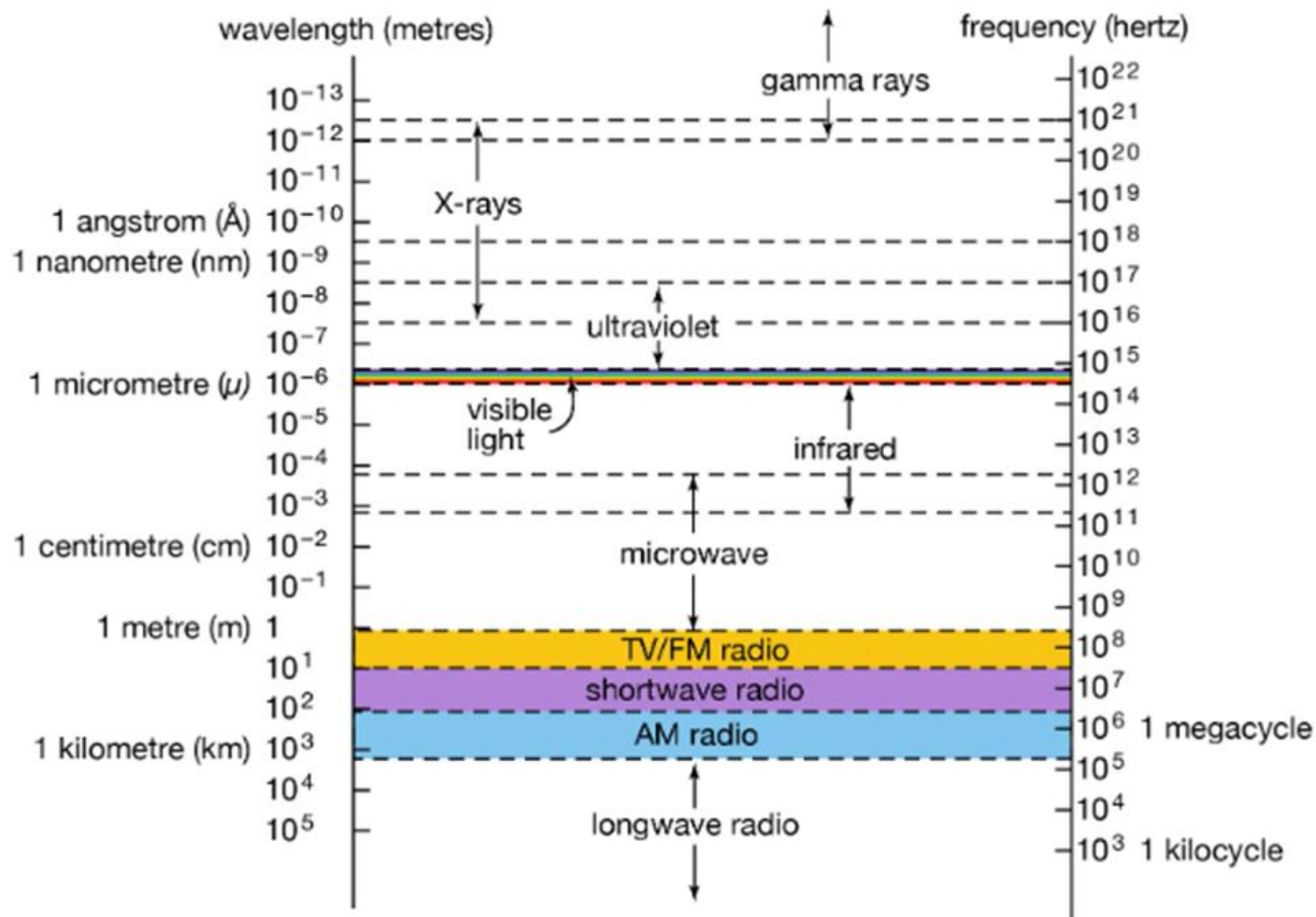
Electromagnetic radiation is the emission and transmission of energy in the form of electromagnetic waves.

Electromagnetic waves travel  $3 \times 10^8$  meters per second (rounded off), or 186,000 miles per second in a vacuum.



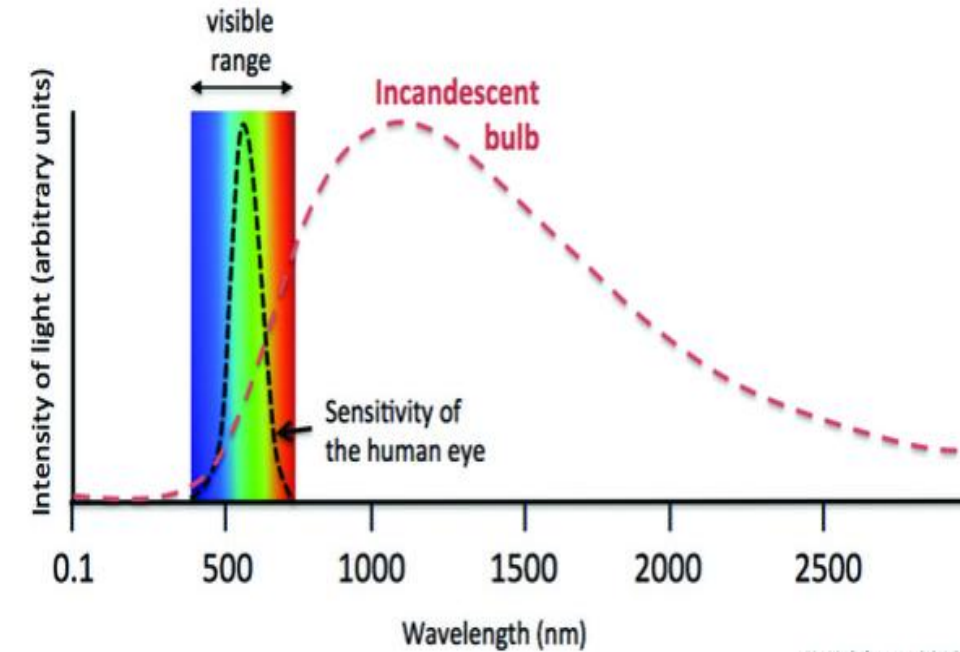
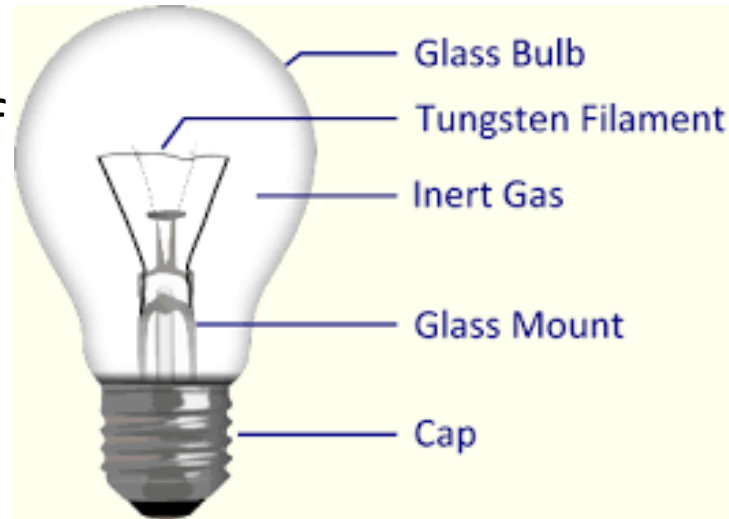
# Types of Electromagnetic Radiation





# Planck's Quantum Theory

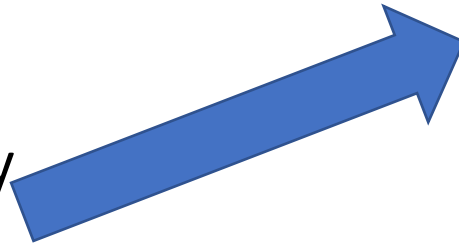
When solids are heated, they emit electromagnetic radiation over a wide range of wavelengths. The dull red glow of an electric heater and the bright white light of a tungsten light bulb are examples of radiation from heated solids.



K. Kelchner, 2012

## In nineteenth century

Amount of radiant energy emitted by an object at a certain temperature depends on its wavelength.



Scientist Could not explain using Established wave and thermodynamic theory

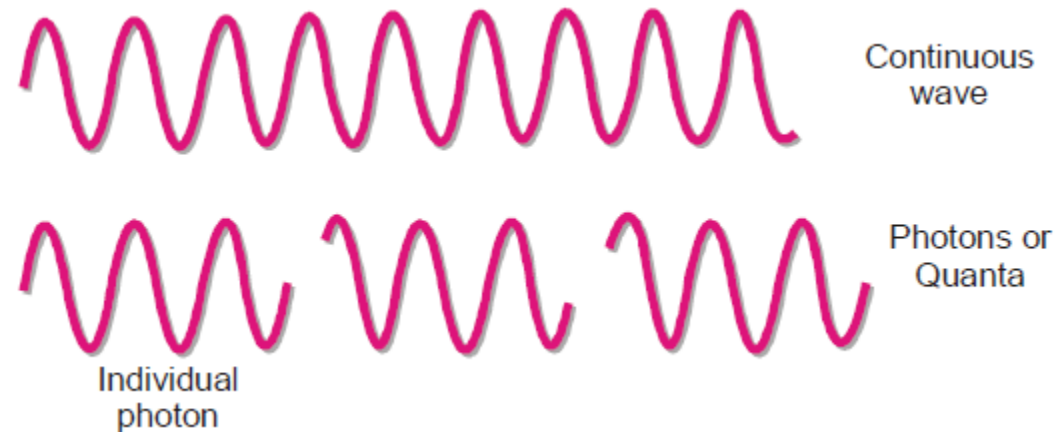


Something fundamental was missing from the laws of classical physics.

# Planck's Quantum Theory

- Planck solved the problem with an assumption that departed drastically from accepted concepts.
- Classical physics assumed that atoms and molecules could emit (or absorb) any arbitrary amount of radiant energy.
- Planck said that atoms and molecules could emit (or absorb) energy only in discrete quantities, like small packages or bundles.
- Planck gave the name quantum to the smallest quantity of energy that can be emitted (or absorbed) in the form of electromagnetic radiation.

1. When atoms or molecules absorb or emit radiant energy, they do so in separate 'units of waves' called quanta or photons. Thus light radiations obtained from energized or 'excited atoms' consist of a stream of photons and not continuous waves.



2. The energy,  $E$ , of a quantum or photon is given by the relation

$$E = h\nu$$

where  $\nu$  is the frequency of the emitted radiation, and  $h$  the Planck's Constant.

$$h = 6.62 \times 10^{-27} \text{ erg sec. or } 6.62 \times 10^{-34} \text{ J sec.}$$

$$c = \lambda \nu$$

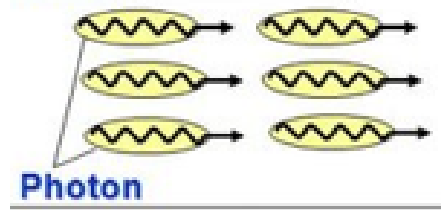
$$E = \frac{hc}{\lambda}$$

3. An atom or molecule can emit (or absorb) either one quantum of energy ( $h\nu$ ) or any whole number multiple of this unit.

Thus radiant energy can be emitted as  $h\nu$ ,  $2h\nu$ ,  $3h\nu$ , and so on, but never as  $1.5 h\nu$ ,  $3.27 h\nu$ ,  $5.9 h\nu$ , or any other fractional value of  $h\nu$  i.e.  $nh\nu$

## Plank`s Quantum theory

Energy of electromagnetic radiation is Quantized (discrete)

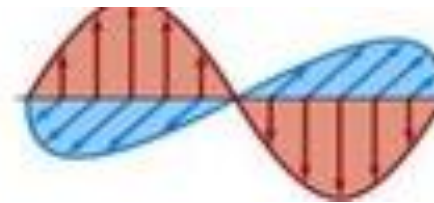


Energy of electromagnetic radiation depends on frequency or wavelength

$$E=hf$$

## Classical theory

Energy of electromagnetic radiation is continuous



Energy of electromagnetic radiation does not depend on frequency or wavelength. Rather it depends on intensity

$$E_{\text{classical}} = k_B T$$

$k_B$  = Boltzmann's constant  
 $T$  = temperature

Calculate the magnitude of the energy of the photon (or quantum) associated with light of wavelength 6057.8 Å. (Å =  $10^{-8}$  cm)

**(a) Calculation of Frequency :**

$$\begin{aligned} \nu &= \frac{c}{\lambda} = \frac{3.0 \times 10^{10} \text{ cm sec}^{-1}}{6057.8 \times 10^{-8} \text{ cm}} \\ &= 4.952 \times 10^{14} \text{ sec}^{-1} \end{aligned}$$

**(b) Calculation of Energy :**

$$\begin{aligned} E &= h\nu = (6.625 \times 10^{-27} \text{ erg sec}) (4.952 \times 10^{14} \text{ sec}^{-1}) \\ &= \mathbf{3.281 \times 10^{-12} \text{ erg}} \end{aligned}$$

The blue color in fireworks is often achieved by heating copper(I) chloride (CuCl) to about 1200°C. Then the compound emits blue light having a wavelength of 450 nm. What is the increment of energy (the quantum) that is emitted at  $4.50 \times 10^2$  nm by CuCl?

The quantum of energy can be calculated from the equation

$$\Delta E = h\nu$$

The frequency  $\nu$  for this case can be calculated as follows:

$$\nu = \frac{c}{\lambda} = \frac{2.9979 \times 10^8 \text{ m/s}}{4.50 \times 10^{-7} \text{ m}} = 6.66 \times 10^{14} \text{ s}^{-1}$$

So

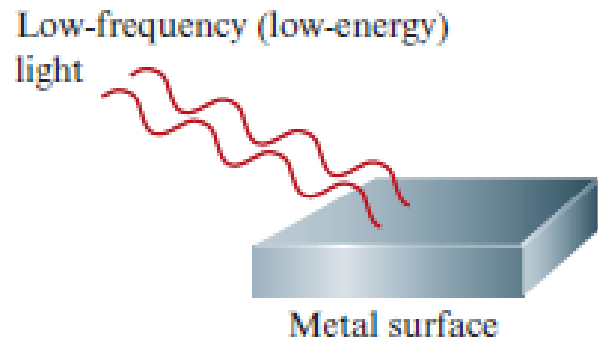
$$\Delta E = h\nu = (6.626 \times 10^{-34} \text{ J} \cdot \text{s})(6.66 \times 10^{14} \text{ s}^{-1}) = 4.41 \times 10^{-19} \text{ J}$$

A sample of CuCl emitting light at 450 nm can lose energy only in increments of  $4.41 \times 10^{-19}$  J, the size of the quantum in this case.

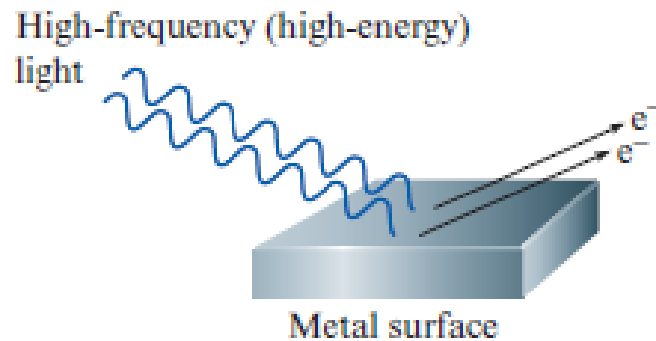
# The Photoelectric Effect

Einstein arrived at this conclusion through his analysis of the photoelectric effect. The photoelectric effect refers to the phenomenon in which electrons are emitted from the surface of a metal when light strikes it. The following observations characterize the photoelectric effect.

1. Studies in which the frequency of the light is varied show that no electrons are emitted by a given metal below a specific threshold frequency,  $\nu_0$ .
2. For light with frequency lower than the threshold frequency, no electrons are emitted regardless of the intensity of the light.
3. For light with frequency greater than the threshold frequency, the number of electrons emitted increases with the intensity of the light.
4. For light with frequency greater than the threshold frequency, the kinetic energy of the emitted electrons increases linearly with the frequency of the light.



a



b

**Figure 7.4** | The photoelectric effect. (a) Light with frequency less than the threshold frequency produces no electrons. (b) Light with frequency higher than the threshold frequency causes electrons to be emitted from the metal.

These observations can be explained by assuming that electromagnetic radiation is quantized (consists of photons), and that the threshold frequency represents the minimum energy required to remove the electron from the metal's surface.

Minimum energy required to remove an electron =  $E_0 = h\nu_0$

Because a photon with energy less than  $E_0$  ( $\nu < \nu_0$ ) cannot remove an electron, light with a frequency less than the threshold frequency produces no electrons (Fig. 7.4). On the other hand, for light where  $\nu > \nu_0$ , the energy in excess of that required to remove the electron is given to the electron as kinetic energy (KE):

$$\text{KE}_{\text{electron}} = \frac{1}{2}mv^2 = h\nu - h\nu_0$$

Mass of electron
Velocity of electron
Energy of incident photon
Energy required to remove electron from metal's surface

$h\nu_0$  is constant for a particular solid and is designated as  $W$ , the work function

$$\frac{1}{2} mv^2 = h\nu - W$$

In a related development, Einstein derived the famous equation

$$E = mc^2$$

in his special theory of relativity published in 1905. The main significance of this equation is that energy has mass. This is more apparent if we rearrange the equation in the following form:

$$m = \frac{E}{c^2}$$

↑  
Mass
 $E \leftarrow$  Energy
←  
Speed of light

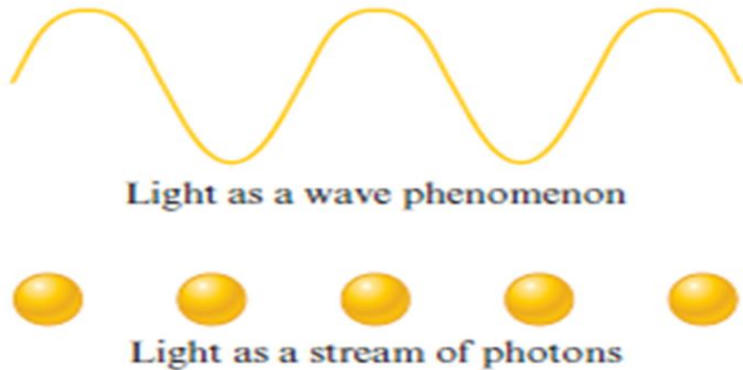
Using this form of the equation, we can calculate the mass associated with a given quantity of energy. For example, we can calculate the apparent mass of a photon. For electromagnetic radiation of wavelength  $\lambda$ , the energy of each photon is given by the expression

$$E_{\text{photon}} = \frac{hc}{\lambda}$$

Then the apparent mass of a photon of light with wavelength  $\lambda$  is given by

$$m = \frac{E}{c^2} = \frac{hc/\lambda}{c^2} = \frac{h}{\lambda c}$$

Note that the *apparent mass* of a photon depends on its wavelength. A photon does not have mass in a classical sense.



**Figure 7.5** | Electromagnetic radiation exhibits wave properties and particulate properties. The energy of each photon of the radiation is related to the wavelength and frequency by the equation  $E_{\text{photon}} = h\nu = hc/\lambda$ .

the important conclusions from the work of Planck and Einstein:

Energy is quantized. It can occur only in discrete units called quanta. Electromagnetic radiation, which was previously thought to exhibit only wave properties, seems to show certain characteristics of particulate matter as well. This phenomenon is sometimes referred to as the dual nature of light.

**1. What is the minimum energy that photons must possess in order to produce photoelectric effect with platinum metal? The threshold frequency for platinum is  $1.3 \times 10^{15} \text{ sec}^{-1}$ .**

The threshold frequency ( $\nu_0$ ) is the lowest frequency that photons may possess to produce the photoelectric effect. The energy corresponding to this frequency is the minimum energy (E).

$$\begin{aligned} E &= h\nu_0 \\ &= (6.625 \times 10^{-27} \text{ erg sec}) (1.3 \times 10^{15} \text{ sec}^{-1}) \\ &= 8.6 \times 10^{-12} \text{ erg} \end{aligned}$$

**2. The work function for metallic cesium is  $3.43 \times 10^{-19} \text{ J}$ . Calculate the kinetic energy and the speed of the electrons ejected by light of 300 nm wavelength. [Hint: Planck's constant, h, is  $6.626 \times 10^{-34} \text{ J s}$ ; the speed of light in a vacuum, c, is  $3.00 \times 10^8 \text{ m/s}$ ; the mass of an electron,  $m_e$ , is  $9.109 \times 10^{-31} \text{ kg}$ ]**

**3. Calculate the kinetic energy of an electron emitted from a surface of potassium metal (work function =  $3.62 \times 10^{-12} \text{ erg}$ ) by light of wavelength  $5.5 \times 10^{-8} \text{ cm}$ .**

1. The work function of cesium metal is  $3.42 \times 10^{-19}$  J. (a) Calculate the minimum frequency of light required to release electrons from the metal. (b) Calculate the kinetic energy of the ejected electron if light of frequency  $1.00 \times 10^{15} \text{ s}^{-1}$  is used for irradiating the metal.
2. The work function of titanium metal is  $6.93 \times 10^{-19}$  J. Calculate the kinetic energy of the ejected electrons if light of frequency  $2.50 \times 10^{15} \text{ s}^{-1}$  is used to irradiate the metal.
3. The work function of potassium is  $3.68 \times 10^{-19}$  J. (a) What is the minimum frequency of light needed to eject electrons from the metal? (b) Calculate the kinetic energy of the ejected electrons when light of frequency equal to  $8.62 \times 10^{15} \text{ s}^{-1}$  is used for irradiation.
4. When light of frequency equal to  $2.11 \times 10^{15} \text{ s}^{-1}$  shines on the surface of gold metal, the kinetic energy of ejected electrons is found to be  $5.83 \times 10^{-19}$  J. What is the work function of gold?
5. A clean metal surface is irradiated with light of three different wavelengths  $\lambda_1$ ,  $\lambda_2$ , and  $\lambda_3$ . The kinetic energies of the ejected electrons are as follows:  $\lambda_1$  :  $2.9 \times 10^{-20}$  J;  $\lambda_2$  : approximately zero;  $\lambda_3$  :  $4.2 \times 10^{-19}$  J. Which light has the shortest wavelength, and which has the longest wavelength?